MH1812 Discrete Mathematics
- Continuous Assessment 1

NTU, AY18/19 S1 20 Sept 2018, 10:30AM - 11:20AM

Name: Tutorial Group: NTU Email:

Note: ***first 3*** *questions are compulsory. Question 4 is bonus, you get the points if you answer it correctly, you don't lose any points if you leave it or answer wrongly. You can write on the back of paper if there is not enough space, leave the tutorial group if you can't remember.*

Question 1 (20 points)

1. Compute 15^{2018} modulo 7 (10 points).

   Solution: Since $15 \equiv 1 \mod 7$, we have $15^{2018} \equiv 1^{2018} \equiv 1 \mod 7$. $\square$

2. Suppose that $Q(x)$ is the statement: $x + 1 = 2x$. What is the truth values of $\forall x \in \mathbb{R}, Q(x)$ and $\exists x \in \mathbb{R}, Q(x)$? (10 points).

   Solution:Since $x + 1 = 2x$ is true if and only if $x = 1$, we see that $Q(x)$ is true if and only if $x = 1$. It follows that $\forall x \in \mathbb{R}, Q(x)$ is false and $\exists x \in \mathbb{R}, Q(x)$ is true. $\square$

Question 2 (50 points)

1. Show that $\neg(p \vee \neg q)$ and $q \wedge \neg p$ are equivalent (30 points)

   (a) using a truth table

   Solution: We have the following truth table

p	q	$\neg q$	$p \vee \neg q$	$\neg(p \vee \neg q)$	$\neg p$	$q \wedge \neg p$
T	T	F	T	F	F	F
T	F	T	T	F	F	F
F	T	F	F	T	T	T
F	F	T	T	F	T	F

   Since the fifth and seventh columns agree, we conclude that $\neg(p \vee \neg q)$ and $q \wedge \neg p$ are equivalent. $\square$

   (b) using logical equivalences

   Solution:

$$\begin{aligned} \neg(p \vee \neg q) &\equiv \neg p \wedge \neg\neg q \quad \text{(De Morgan)} \\ &\equiv \neg p \wedge q \quad \text{(Double Negation)} \\ &\equiv q \wedge \neg p \quad \text{(Commutative)} \end{aligned}$$

   We conclude $\neg(p \vee \neg q)$ and $q \wedge \neg p$ are equivalent. $\square$

2. Decide whether the following argument is valid (20 points):

$$\begin{aligned} &p \to r \\ &q \to r \\ &q \vee \neg r \\ &\therefore \neg p \end{aligned}$$

Solution:

We have the following truth table

p	q	r	$p \to r$	$q \to r$	$q \vee \neg r$	$\neg p$
F	F	F	T	T	T	T
F	F	T	T	T	F	T
F	T	F	T	F	T	T
F	T	T	T	T	T	T
T	F	F	F	T	T	F
T	F	T	T	T	F	F
T	T	F	F	F	T	F
T	T	T	T	T	T	F

Since the last row is critical such that all premises are true, the conclusion is false, therefore the argument is invalid.

Counter-example: p is true, q is true, and r is true.

$\square$

Question 3 (30 points)

Use Mathematical Induction to prove that $1 + 2^n \leq 3^n$ for all integers $n \geq 1$.

Solution:Let $P(n) : 1 + 2^n \leq 3^n$.

$P(1) : 1+2^1 \leq 3^1$, which is true as both sides are equal to 3. Assume $P(k)$ is true: $1+2^k \leq 3^k$, We show that $P(k+1)$ is also true.

$$1 + 2^{k+1} = (1 + 2^k) + 2^k \leq 3^k + 3^k = 2 \cdot 3^k < 3 \cdot 3^k = 3^{k+1}.$$

$\square$

Question 4 (Bonus Question) (10 points)
Prove or disprove: there is a prime $p > 3$ such that $p, p+2$, and $p+4$ are prime.

Solution:The statement is not true. $p \mod 3, p+2 \mod 3, p+4 \mod 3$ are distinct. One of them must be $0 \mod 3$. It follows that one of them must be divisible by 3, so it cannot be a prime. $\square$